

Data Engineering for AI Applications

Session Agenda

Duration: 3 Hours

Objective

Build strong data-centric thinking for AI system development. This session moves beyond model architecture to focus on the discipline that determines whether an AI system is trustworthy: the quality, structure, and pipelines behind its data — from raw file formats through cleaning, text preprocessing, annotation, and dataset governance.

Topics Covered

- Structured vs unstructured data
- ETL fundamentals
- Data cleaning pipelines
- Text preprocessing
- File formats (PDF, DOCX, HTML, CSV, JSON)
- OCR workflow awareness
- Annotation fundamentals
- Dataset quality assessment
- Dataset versioning
- SQL basics
- Data privacy awareness
- Data governance fundamentals

Learning Outcome

Learners understand how data quality, structure, and pipelines affect downstream AI system performance.

Detailed Session Schedule

Duration	Topic / Content
Part 1 — Data Foundations & Cleaning (85 min)	
10 min	Welcome & Why Data-Centric Thinking Matters: Framing the session — why data quality and structure, not just model architecture, determine AI system performance.
15 min	Structured, Semi-Structured & Unstructured Data: The three tiers of data organization, and why each demands a different processing approach.
15 min	File Formats Deep Dive: CSV, JSON, PDF, DOCX, and HTML — structure, parsing challenges, and choosing the right format for the task.
20 min	ETL Fundamentals + Hands-on Demo: Extract → Transform → Load explained and demonstrated; batch vs streaming; common tooling. Standalone demo: clean a raw CSV end-to-end.
20 min	Data Cleaning Pipelines + Hands-on Demo: Common data issues, missing-value strategies, and outlier detection. Standalone demo: a repeatable Profile → Clean → Validate → Log pipeline.
5 min	Quick Recap & Check-in: Consolidate Part 1 concepts and clear doubts before the break.
10 min	Break
Part 2 — Text, Datasets & Responsible Data Practice (85 min)	
15 min	Text Preprocessing + Hands-on Demo: Tokenization, normalization, stopword removal, and stemming vs lemmatization. Standalone demo: a full preprocessing pipeline on raw feedback text.
10 min	OCR Workflow Awareness: How scanned documents and images become machine-readable text, and where OCR commonly breaks down.
10 min	Annotation Fundamentals: Labeling data for supervised learning — annotation types, and why consistency and inter-annotator agreement matter.
15 min	Data Quality Deep Dive: Missing values, noise, bias, and hallucination risk — with a practical data quality checklist and before/after examples.
5 min	Dataset Versioning: Why datasets need version control, and basic practices for tracking dataset changes over time.
20 min	SQL Basics + Hands-on Demo: Core clauses through subqueries — SELECT, WHERE, JOIN (including JOIN types), GROUP BY, HAVING, ORDER BY/LIMIT. Standalone demo: querying a sample dataset for a real business question.

Duration	Topic / Content
5 min	Data Privacy & Governance Fundamentals: PII and anonymization basics, plus data ownership, access control, and compliance fundamentals.
5 min	Recap & Key Takeaways: Tie all 12 topics back to the session's learning outcome; final Q&A.